



GUJARAT TECHNOLOGICAL UNIVERSITY

Shree Swami Atmanand Saraswati Institute of Technology

A

Project Report on

Analysis of Global Tech & AI Startups Using Python

Under subject of

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COMPUTER ENGINEERING

Under the Guidance of

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CERTIFICATE

This is to certify that the project entitled “Analysis of Global Tech & AI Startups” has been prepared by Gajera Diya (230760107030), Bhungaliya Jensi (240760107020) and Diyora Khushi (240760107046), in the subject of Python for Data Science (BE05000231), under my guidance, in fulfilment of the degree of Bachelor of Engineering in Computer Engineering (5th Semester) of Gujarat Technological University, Ahmedabad, during the academic year 2026-2027.

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ABSTRACT

This microproject presents a Python-based analysis of the Global Tech & AI Startups Dataset (2020–2026), containing 25,000 technology and AI startups sourced from Kaggle. The study applies Data Science techniques including data cleaning, exploratory data analysis, statistical correlation, and visualization using Pandas, NumPy, Matplotlib, and Seaborn. The analysis investigates geographic and sectoral composition, funding trends, domain-wise funding and valuation, the relationship between funding and valuation, company size across funding stages, and layoff patterns.

Data quality issues, including missing values and a duplicate identifier, were addressed before analysis. The findings indicate that the dataset is highly concentrated in the USA and the Generative AI domain. A strong positive correlation was observed between funding and valuation (Pearson $r = 0.945$). Company size also increases with funding stage, while Web3/Crypto shows a comparatively higher layoff rate. Overall, the project demonstrates an evidence-based approach to understanding global technology and AI startup trends using Python.

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CHAPTER 1 : INTRODUCTION

1.1 Background

The global startup ecosystem, particularly in technology and artificial intelligence (AI), has seen rapid change between 2020 and 2026 — a period marked by a post-pandemic funding boom, a subsequent slowdown often called the “funding winter,” and a wave of AI-driven company creation following the rise of generative AI.

This microproject applies core Python for Data Science techniques to a real-world dataset of 25,000 technology and AI startups in order to explore, clean, and analyse patterns in startup funding, growth, and outcomes.

1.2 Purpose of the Project

The project was carried out as part of the “Python for Data Science” subject, with an emphasis on demonstrating practical skills in Pandas, NumPy, data cleaning, exploratory data analysis (EDA), descriptive statistics, data aggregation, and data visualization using Matplotlib and Seaborn, rather than building predictive machine-learning models.

1.3 Importance of the Topic

Raw startup data, on its own, is difficult to interpret. Understanding funding, valuation, and workforce patterns across thousands of companies requires a systematic, evidence-based analytical approach rather than intuition. Turning large, raw datasets into clear, defensible insights is a core skill for any Computer Engineering student, with direct relevance to analytics, finance, and business-intelligence roles.

CHAPTER 2 : PROPOSAL

2.1 Problem Statement

Raw startup data, on its own, is difficult to interpret — 25,000 rows of numbers and categories do not, by themselves, tell a story. Without systematic cleaning and analysis, it is not possible to answer basic but important questions such as: which countries and sectors dominate the startup landscape, whether funding actually predicts a company's valuation, whether later-stage companies are genuinely bigger and more valuable, and which sectors have been most affected by layoffs. This project addresses the problem of turning a large, raw, real-world dataset into clear, evidence-based answers to these questions using Python.

2.2 Aim and Objectives

The objectives of this microproject are to:

- Inspect and understand the structure, quality, and limitations of the Global Tech & AI Startups dataset.
- Clean the dataset by identifying and resolving data quality issues (missing values and a duplicate identifier) using clearly justified, documented decisions.
- Explore the dataset using Pandas and NumPy to compute descriptive statistics, grouped aggregations, and correlations.
- Visualize key patterns using Matplotlib and Seaborn, choosing chart types appropriate to each question.
- Answer six specific research questions covering geographic/sectoral composition, funding trends over time, funding and valuation by domain, the funding-valuation relationship, the effect of funding stage on company size, and layoff patterns.
- Draw meaningful, evidence-based conclusions strictly from the dataset itself, without introducing unsupported claims.

2.3 Scope of the Project

The scope of this microproject is limited to descriptive analysis and visualization of the given dataset using Python. It does not include building a predictive machine-learning model, deploying a live dashboard, or collecting new data beyond the provided Kaggle dataset. Extensions of this kind are discussed as future scope in Chapter 5.

2.4 Tools and Technologies

Tool / Library	Purpose
Python 3	Core programming language

Tool / Library	Purpose
Jupyter Notebook / Google Colab	Interactive development environment for analysis
Pandas	Loading, cleaning, grouping, and aggregating tabular data
NumPy	Numerical operations supporting Pandas computations
Matplotlib	Base plotting library for charts (bar, line, scatter)
Seaborn	Statistical visualization (boxplots, heatmaps) built on Matplotlib

Table 2.1 : Tools and Technologies Used

2.5 Methodology

The project followed a standard Data Science workflow, carried out step by step in a Jupyter notebook:

Stage	Activity	Purpose
1	Data Loading	Load the CSV file into a Pandas DataFrame.
2	Understanding the Dataset	Inspect dataset structure, data types, missing values, duplicates, and unique category counts before any changes.
3	Data Cleaning	Resolve identified issues such as missing values and duplicate IDs using documented and justified decisions; save a cleaned CSV file.
4	Exploratory Data Analysis	Explore the cleaned data question by question using grouping, aggregation, and correlation analysis.
5	Data Visualization	Visualize each question's findings using an appropriate chart type.
6	Synthesis	Summarize and interpret the key findings obtained from the analysis.

Table 2.2 : Project Methodology – Stages and Purpose

CHAPTER 3 : DATASET AND DATA PREPARATION

3.1 Dataset Description

The dataset used is the “Global Tech & AI Startups Dataset (2020–2026)” sourced from Kaggle. It contains 25,000 rows (one per startup company) and 17 columns, covering company identity, geography, sector, funding, valuation, financial health, workforce, and outcome status.

Column	Description	Type
Company_ID	Unique identifier code for each startup	Text
Domain	Industry/sector the startup operates in	Categorical (10)
Founding_Year	Year the company was founded (2012–2025)	Integer
Country	Country of headquarters	Categorical (10)
City	City of headquarters	Categorical (29)
Funding_Stage	Current investment stage (Seed, Series A/B/C+, Pre-IPO, Post-IPO)	Categorical (6)
Total_Funding_USD_Millions	Total money raised to date (USD millions)	Float
Valuation_USD_Millions	Company's current valuation (USD millions)	Float
Revenue_ARR_Millions	Annual Recurring Revenue (USD millions)	Float
Monthly_Burn_Rate_Millions	Monthly cash spend (USD millions)	Float
Runway_Months_2024	Months of cash remaining as of 2024	Float
Peak_Headcount_2023	Highest employee count in 2023	Integer
Layoffs_2024_2025	Number of employees laid off, 2024–2025	Integer
Current_Headcount_2026	Employee count as of 2026	Integer
Investor_Tier	Quality/tier of backing investors	Categorical (5)
AI_Adoption_Level	Company's internal AI adoption (Low/Medium/High)	Categorical (3) + missing
Acquisition_Status	Company outcome (Independent, Acquired, Closed, IPO, etc.)	Categorical (5)

Table 3.1 : Column Descriptions of the Dataset

3.2 Data Quality Summary

Inspection of the raw dataset revealed the following:

- Shape: 25,000 rows × 17 columns, with no fully duplicate rows.

- Missing values: only `AI_Adoption_Level` had missing entries — 2,592 rows (about 10.4% of the dataset). Every other column was fully populated.
- Duplicate identifiers: exactly one `Company_ID` (“7791E581”) was shared by two different companies with different domains, countries, and financial figures — an ID collision rather than a duplicate record.
- Logical consistency: `Current_Headcount_2026` was found to equal `Peak_Headcount_2023` minus `Layoffs_2024_2025` for all 25,000 rows, `City` values mapped consistently to a single `Country` each, and there were no negative values in any numeric column — indicating that the dataset is internally consistent and reliable.

3.3 Data Cleaning

Cleaning was limited to genuine, identified problems — no changes were made that were not justified by the data-quality inspection above.

Handling the duplicate `Company_ID`: the two rows sharing the ID “7791E581” were confirmed to be different, real companies, so neither row was deleted. Instead, a new unique 8-character ID (matching the existing format) was generated for the second occurrence, using a fixed random seed (42) for reproducibility.

Handling missing `AI_Adoption_Level` values: the 2,592 missing values were filled with the label “Unknown” rather than deleted or imputed with a guessed value, to avoid discarding about 10% of the dataset, avoid fabricating information, and remain fully transparent about what is and is not known.

Additional cleaning: all text/categorical columns were stripped of leading and trailing whitespace to prevent silent duplicate categories (e.g., “India” vs. “India ”).

Verification: after cleaning, 0 missing values and 0 duplicate `Company_ID` values remained, and all 25,000 original rows were retained — confirming that cleaning fixed the identified problems without discarding any data.

CHAPTER 4 : ANALYSIS AND RESULTS

4.1 Exploratory Data Analysis – Question-wise Results

Six research questions (Q1–Q6) were investigated using the cleaned dataset. Each question below states its purpose, the Pandas/statistical method used, and the key numerical results. The corresponding charts are presented in Section 4.2.

Q1. Which countries and domains have the most startups?

Method: `value_counts()` on the Country and Domain columns, converted to percentages.

Top Countries	Share	Top Domains	Share
USA (11,178)	44.7%	Generative AI (4,958)	19.8%
China (3,755)	15.0%	HealthTech (3,760)	15.0%
India (2,560)	10.2%	FinTech (3,060)	12.2%
UK (1,998)	8.0%	ClimateTech (2,507)	10.0%

Table 4.1 : Top Countries and Domains by Startup Count (Q1)

Q2. How has startup founding activity trended over time (2012–2025)?

Method: `groupby('Founding_Year').size()`, examined as a year-over-year trend. **Result:** founding activity rose overall from 492 companies in 2012 to a peak of 5,047 companies in 2020, then declined every year afterward, down to 257 companies in 2025 — a pattern consistent with either a genuine post-2020 slowdown or a data-recency effect; the dataset alone cannot distinguish between these explanations.

Q3. Which domains attract the highest funding and valuation?

Method: grouped mean and median of `Total_Funding_USD_Millions` and `Valuation_USD_Millions` by Domain.

Domain	Median Funding (\$M)	Median Valuation (\$M)	Mean Valuation (\$M)
Generative AI	18.6	143.2	885.8
SpaceTech	14.3	113.1	747.9
Web3/Crypto	8.5	72.1	402.6
Cybersecurity	5.8	53.6	292.3
EdTech	5.9	51.6	298.7

Table 4.2 : Median Funding, Median Valuation and Mean Valuation by Domain (Q3)

Note: five of the ten domains are shown (the top three, plus EdTech and Cybersecurity, which have the lowest median funding). Median funding for all ten domains is shown in Figure 4.4.

Result: Generative AI leads on both funding and valuation; SpaceTech is a strong second. The gap between mean and median valuation for every domain confirms a right-skewed distribution, where a small number of highly funded companies pull the average well above the typical company.

Q4. Is there a relationship between total funding raised and valuation?

Method: Pearson correlation coefficient between Total_Funding_USD_Millions and Valuation_USD_Millions, plus a full correlation matrix across numeric columns. **Result:** Pearson $r = 0.945$ — a very strong positive relationship. Monthly_Burn_Rate_Millions and Peak_Headcount_2023 are nearly perfectly correlated ($r = 0.98$), while Runway_Months_2024 is only weakly related to the other variables ($r = 0.05$ – 0.32). Correlation shows that the variables move together, not that one causes the other.

Q5. How does funding stage relate to valuation, revenue, and headcount?

Method: Funding_Stage was set as an ordered categorical column (Seed → Series A → Series B → Series C+ → Pre-IPO → Post-IPO), then median Valuation, Revenue (ARR), and Headcount were compared across stages.

Funding Stage	Median Valuation (\$M)	Median Revenue ARR (\$M)	Median Headcount
Seed (8,859 cos.)	25.5	1.6	23
Series A (6,154)	70.6	4.7	37
Series B (5,014)	187.4	11.5	78
Series C+ (2,996)	480.6	30.4	199
Pre-IPO (1,257)	1,233.4	77.8	451
Post-IPO (720)	2,950.3	193.7	1,093

Table 4.3 : Median Valuation, Revenue and Headcount by Funding Stage (Q5)

Result: a clean, consistent step-up pattern held across all three metrics from Seed to Post-IPO, matching real-world startup growth expectations.

Q6. Which domains/countries had the highest layoff rates in 2024–2025?

Method: a derived column, Layoff_Rate_Pct = $(\text{Layoffs}_{2024_2025} / \text{Peak_Headcount}_{2023}) \times 100$, was created and averaged by Domain and by Country. **Result:** Web3/Crypto had a dramatically higher average layoff rate (56.8%) than every other domain (7.4%–15.2%) — about 3.7 times the next-highest domain (EdTech, 15.2%). Generative AI had the lowest layoff rate (7.4%). Differences by country were small (15.6%–16.8%). Overall, 61.2% of companies reported zero layoffs in 2024–2025.

4.2 Data Visualization

For every chart below, what it shows, why that chart type was chosen, and the insight it reveals are explained alongside the figure.

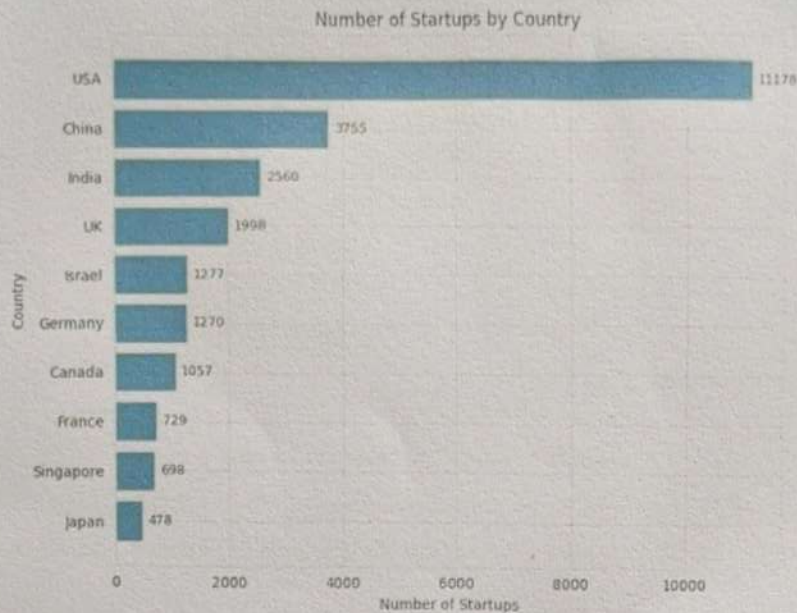


Figure 4.1 : Number of Startups by Country

A horizontal bar chart was chosen because country names vary in length and sorting by value lets the most represented country stand out. Insight: the USA dominates with 44.7% of all startups, more than the next five countries combined — a key caveat for interpreting every later result.

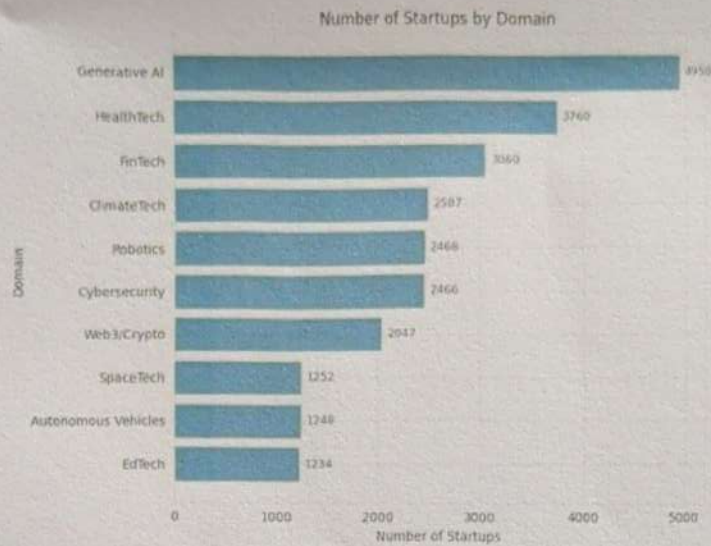


Figure 4.2 : Number of Startups by Domain

Same horizontal bar format, for consistency across the 10 categories. Insight: Generative AI is the largest single domain (19.8%), and the top three domains account for 47% of the entire dataset.

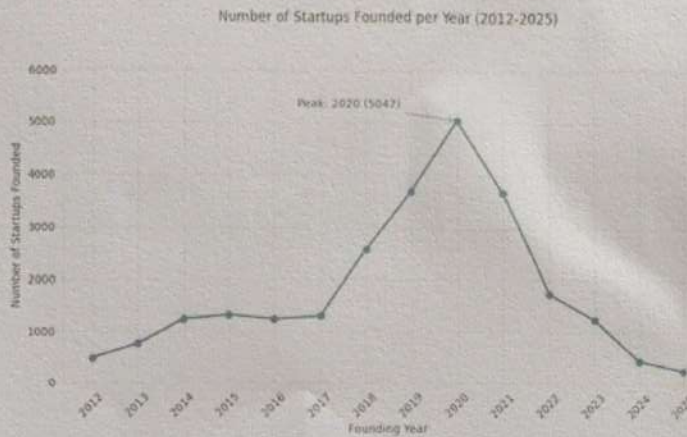


Figure 4.3 : Number of Startups Founded per Year, 2012–2025

A line chart suits time-ordered data, emphasising the direction and shape of the trend. Insight: founding activity rose to a peak of 5,047 companies in 2020, then declined continuously through 2025 (257 companies).

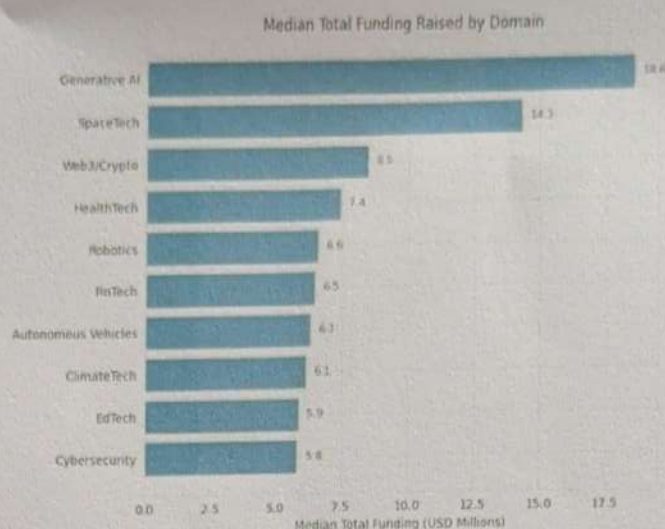


Figure 4.4 : Median Total Funding Raised by Domain

The median (rather than the mean) represents a “typical” company without outlier distortion. Insight: Generative AI’s median funding (\$18.6M) is more than double that of the third-ranked domain (Web3/Crypto, \$8.5M).

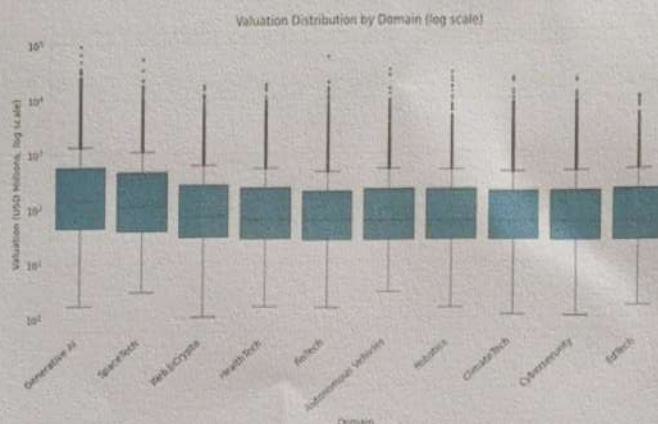


Figure 4.5 : Distribution of Company Valuation by Domain (Log Scale)

A boxplot on a log scale reveals skew and outliers that a bar chart of averages would hide, since valuations range from about \$1M to over \$91,000M. Insight: every domain has a long tail of high-value outliers; Generative AI and SpaceTech have the highest overall valuation distributions.

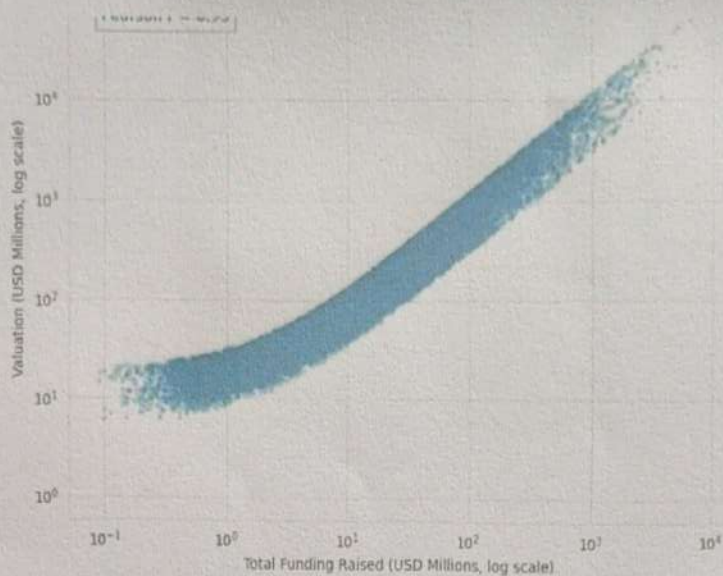


Figure 4.6 : Total Funding vs. Valuation (Log-Log Scale)

A scatter plot visualises the relationship between two numeric variables; both axes were log-scaled, and transparency (25% opacity) reveals density among the 25,000 overlapping points. Insight: the points form a tight upward band, confirming the very strong positive correlation ($r = 0.945$).

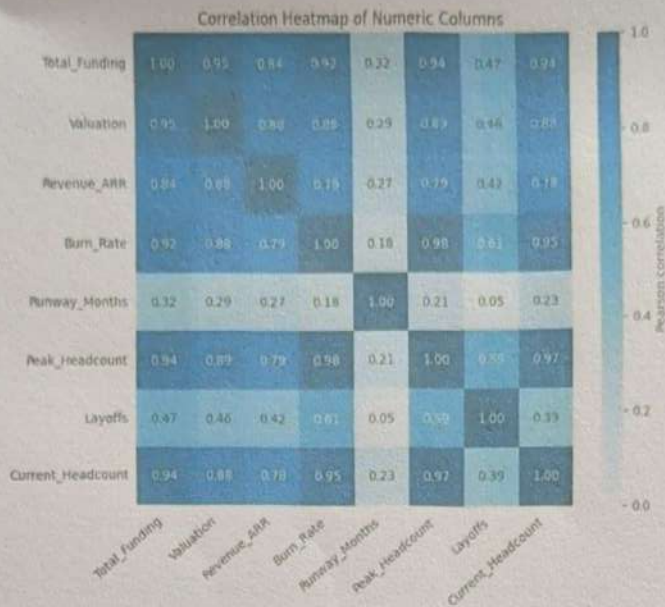


Figure 4.7 : Correlation Heatmap Across Numeric Columns

A heatmap allows many pairwise relationships to be scanned at once. Insight: burn rate and peak headcount are nearly perfectly correlated (0.98), while runway is only weakly related to any other variable (0.05–0.32).

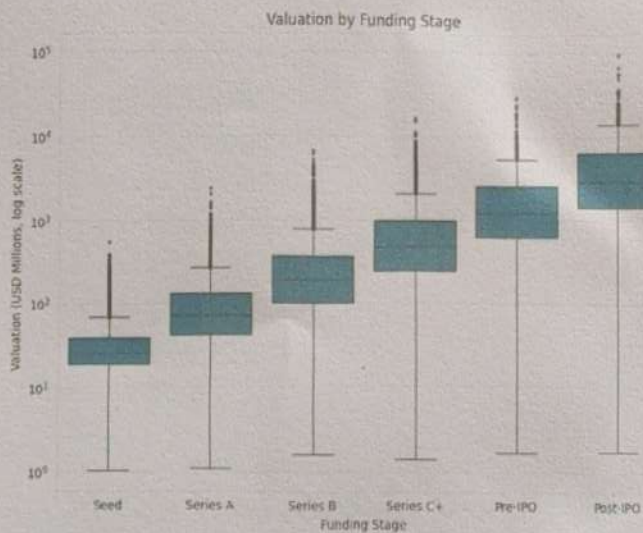


Figure 4.8 : Valuation Distribution by Funding Stage

A boxplot on a log scale shows the full distribution at each stage, with stages ordered in their real business sequence. Insight: valuation rises in a clear, consistent step at every stage, from a median of \$25.5M at Seed to \$2,950.3M at Post-IPO.

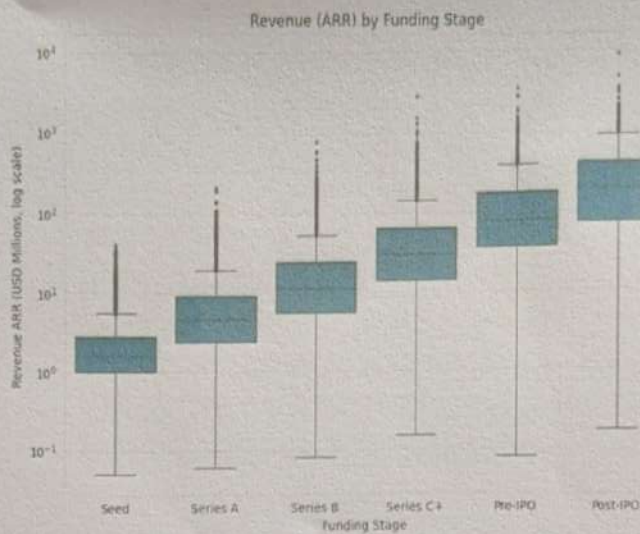


Figure 4.9 : Revenue (ARR) Distribution by Funding Stage

Same chart type and ordering, applied to revenue instead of valuation, for direct comparison. Insight: revenue follows the same clean step-up pattern (median \$1.6M at Seed to \$193.7M at Post-IPO).

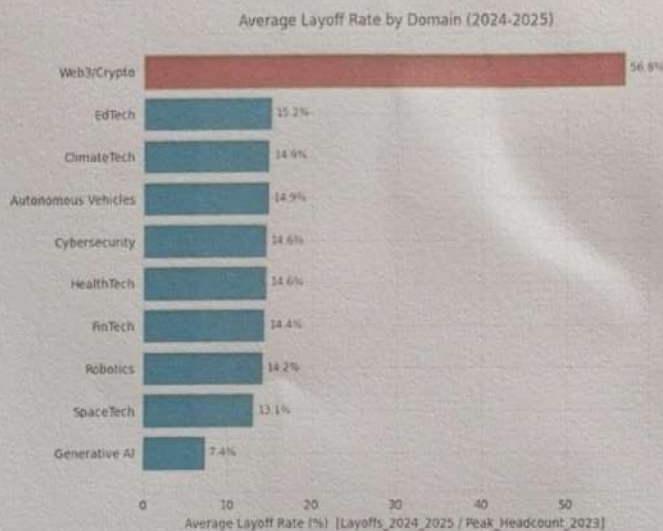


Figure 4.10 : Average Layoff Rate by Domain, 2024–2025

A horizontal bar chart, sorted by value with the most extreme domain highlighted, draws attention to the standout finding. Insight: Web3/Crypto’s average layoff rate (56.8%) is about 3.7 times that of the next-highest domain (EdTech, 15.2%), while Generative AI has the lowest (7.4%).

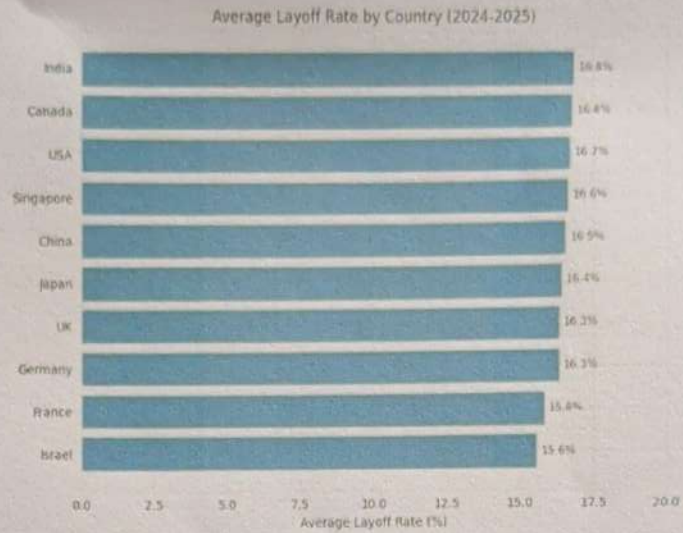


Figure 4.11 : Average Layoff Rate by Country, 2024–2025

Identical format, for fair visual comparison. Insight: country-level differences are small (15.6%–16.8%) and do not point to a meaningful pattern.

CHAPTER 5 : KEY FINDINGS AND IMPACT

5.1 Key Findings

1. The dataset is concentrated in a few countries and domains (Q1).

The USA accounts for 44.7% of all startups; Generative AI accounts for 19.8%, and the top three domains together make up 47% of the dataset.

2. Startup founding peaked in 2020, then declined sharply through 2025 (Q2).

From a peak of 5,047 companies founded in 2020, the count fell every year to just 257 in 2025 — a trend that may reflect a real slowdown, incomplete recent data, or both.

3. Generative AI leads in funding and valuation, not just company count (Q3).

It has the highest median funding (\$18.6M) and the highest median (\$143.2M) and mean (\$885.8M) valuation of any domain; SpaceTech is a strong second.

4. Funding and valuation are very strongly correlated (Q4).

Pearson $r = 0.945$. Burn rate and peak headcount are nearly perfectly correlated ($r = 0.98$); runway is only weakly related to the other variables ($r = 0.05$ – 0.32).

5. Later funding stages consistently mean bigger, more valuable, higher-revenue companies (Q5).

Median valuation, revenue, and headcount all rise cleanly and consistently from Seed through Post-IPO.

6. Web3/Crypto had a dramatically higher layoff rate than every other domain (Q6).

Its average layoff rate was 56.8%, versus 7.4%–15.2% for the other domains; Generative AI had the lowest at 7.4%. Country-level differences were negligible, and 61.2% of companies had zero layoffs.

Cross-cutting observation: Generative AI and Web3/Crypto repeatedly appear as opposite extremes across independent questions — Generative AI performs best (highest funding, highest valuation, lowest layoff rate), while Web3/Crypto shows the most instability (moderate funding and valuation, but by far the highest layoff rate). A pattern that reappears across more than one independent analysis is more trustworthy than one that appears only once.

5.2 Learning Outcomes

This microproject strengthened practical skills in the Python Data Science stack — Pandas, NumPy, Matplotlib, and Seaborn — and, more importantly, in the discipline of justifying every data-cleaning decision rather than editing data arbitrarily. It also reinforced the

difference between correlation and causation, and the value of visual exploration alongside numerical summaries.

5.3 Challenges and Limitations

The dataset's heavy skew toward the USA and Generative AI limits how far the findings generalise to the global startup ecosystem. The post-2020 decline in founding activity is ambiguous and could reflect either a genuine slowdown or incomplete data for the most recent years — the dataset alone cannot resolve this. All findings are descriptive and correlational, not causal.

5.4 Future Scope

- Statistical significance testing (e.g., ANOVA or a t-test) could confirm whether the layoff-rate difference for Web3/Crypto (Q6) and the funding/valuation differences between domains (Q3) are statistically significant rather than only descriptively large.
- The relationship between `AI_Adoption_Level` and `Acquisition_Status` was not examined in this project and could be explored in full, including a deeper look at why `AI_Adoption_Level` was unreported for about 10% of companies.
- If a data-collection timestamp were available, the post-2020 decline in founding activity (Q2) could be distinguished as a genuine trend versus a data-recency artifact.
- A simple predictive model (e.g., logistic regression) could be explored as a stretch goal to estimate acquisition outcome from funding and growth metrics.
- The analysis could be extended into an interactive dashboard (e.g., using Streamlit or Power BI) for dynamic filtering and exploration.

CONCLUSION

This microproject analysed the Global Tech & AI Startups Dataset (2020–2026), a dataset of 25,000 startups across 10 countries and 10 technology domains, using Pandas, NumPy, Matplotlib, and Seaborn. The workflow followed standard Data Science practice: understanding the raw data, cleaning it through documented and justified decisions, and answering six analysis questions using grouping, aggregation, correlation, and visualization.

Across the six questions, a few themes stood out. The dataset is not evenly distributed — it leans heavily toward the USA and toward Generative AI — so the findings describe this dataset specifically rather than the global startup ecosystem as a whole. Funding, valuation, revenue, and headcount move together in the way real startup economics would predict, giving confidence that the dataset is internally consistent. Two domains, Generative AI and Web3/Crypto, repeatedly stood out as opposite extremes across multiple independent questions — a more convincing pattern than any single statistic alone. This analysis is descriptive, not causal: correlations show that variables move together, not that one causes another.

Overall, this project demonstrates the core Python for Data Science toolkit expected at a microproject level — loading and inspecting data, cleaning it with justified decisions, exploring it through aggregation and correlation, and communicating findings through purposefully chosen visualizations — grounded strictly in what the dataset actually shows.

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- Python Software Foundation. Python 3 Documentation. <https://docs.python.org/3/>